Exercises

An irreducible polynomial is assumed to be nonzero and not a unit. Every field is nonzero. These exercises were designed for this course. A line in finite geometry is an incidence block and need not appear straight in a drawing over the real numbers.

1. Determine whether $p = x^3 + x + 1$ is irreducible in $\mathbb{F}_2[x]$. List every element of $\mathbb{F}_2[x]/(p)$. Compute $[x^2]^{-1}$ by Euclidean division and back-substitution, and check it by multiplication.
2. In $GF(4) = \mathbb{F}_2[\alpha]$, where $\alpha^2 = \alpha + 1$, calculate $(\alpha + 1)^3$, $(\alpha + 1)^2 + \alpha$, and all nonzero inverses. Use additive element orders to prove that this ring is not isomorphic to $\mathbb{Z}_4$.
3. Construct the cyclic Latin square of order 3 and determine whether it is reduced. Construct $L_2(x, y) = 2x + y \pmod 3$. Compare its orthogonality with the first square, first-row normalization, and reducedness.
4. Derive v, b, k, r and the number of directions for $AG(2, 3)$. Find the line through $(1, 2), (2, 1)$ and list all its points. Then find the parallel line through $(1, 1)$ and list its points.
5. Use double counting, without quoting projective plane parameters, to find b, r for a $2 - (7, 3, 1)$ design. Check the divisibility conditions for $2 - (16, 6, 1)$, and prove that this design still cannot exist.
6. In the subspace model of $PG(2, q)$, prove that two distinct points determine a unique line and that two distinct lines meet in exactly one point. Derive the number of points on each line, lines through each point, and total points and lines. Explain what is being grouped each time you divide by $q - 1$.
7. In $GF(9) = \mathbb{F}_3[\alpha]$, where $\alpha^2 = 2$, prove that α does not generate the multiplicative group but $1 + \alpha$ does. List all eight powers of $1 + \alpha$ and find its inverse.
8. Prove that $f = x^4 + x + 1 \in \mathbb{F}_2[x]$ is irreducible. Checking only that 0 and 1 are not roots is insufficient. Exclude quadratic factors. Then prove that $[x]$ has multiplicative order 15 in the quotient field.
9. Over $\mathbb{Z}_4$, compare $L_1(x, y) = x + y$ and $L_3(x, y) = 3x + y$. Is each square Latin? Are they orthogonal? Give two different cells with the same ordered pair, and identify the operation that fails in the field proof.
10. Construct three MOLS over $GF(4)$ using $L_a(x, y) = ax + y$. List the first column of each square, with rows ordered $0, 1, \alpha, \alpha + 1$. Find the unique cell giving the ordered pair $(1, \alpha)$ in $(L_\alpha, L_{\alpha+1})$. Use the general equations to explain orthogonality.

Exercises (continued)

11. For the Fano blocks $123, 145, 167, 246, 257, 347, 356$, verify the replication of each point and the coverage of every pair. After deleting line 123, list the three affine parallel classes. Translate line 145 into codeword positions for the matrix H in Chapter 16, and write the seven-bit word.
12. Let M be the 7×7 real point-block incidence matrix of the Fano plane in Exercise 11. Find MM^T, $|\det M|$, and a formula for M^{-1}. Why is the same 0/1 matrix singular over $\mathbb{F}_2$? Explain why the real-rank proof does not transfer directly.
13. Suppose finite fields satisfy $E \subseteq K$, with $|E| = p^d$ and $|K| = p^n$. Use products of basis elements to prove $d \mid n$. Decide whether $GF(4)$ can embed into $GF(8)$. Comparing set sizes alone is insufficient.
14. Let $K = GF(16)$. Prove that $S = \{a \in K : a^4 = a\}$ has exactly four elements and is a subfield. Count its roots using the cyclic structure of $K^\times$, and verify closure under addition, multiplication, and inverses.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study